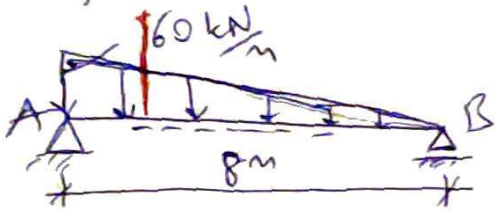
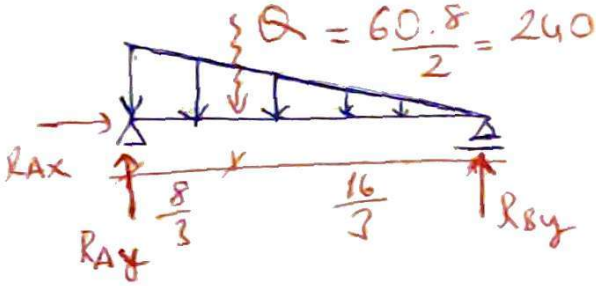
Support Reactions

Examples

1)





$\Sigma P_x = 0 \Rightarrow R_{Ax} = 0$

$\Sigma M_B = 0 \Rightarrow$

$R_{Ay} \cdot 8 - Q \cdot \frac{2}{3} 8 = 0$

$R_{Ay} = \frac{240 \cdot 2 \frac{8}{3}}{8} = 160\ kN$

$\Sigma M_A = 0 \Rightarrow$

$R_{By} \cdot 8 - Q \cdot \frac{8}{3} = 0$

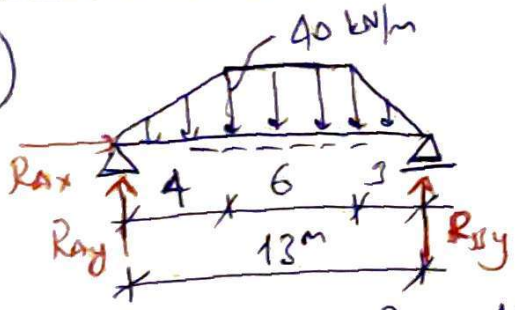
$R_{By} = \frac{240 \cdot 8/3}{8} = 80\ kN$

Kontrol: $\Sigma P_y = 0$

$Q - R_{Ay} - R_{By} = 0$ ✓

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2)



$\Sigma P_x = 0 \Rightarrow R_{Ax} = 0$

$\Sigma M_B = 0 \Rightarrow R_{Ay} \cdot 13 - 40 \frac{4}{2} \cdot \left(3 + \frac{4}{3}\right) - 40 \cdot 6 \cdot \left(3 + \frac{6}{2}\right) - 40 \frac{3}{2} \cdot \left(\frac{2}{3} \cdot 3\right) = 0$

$R_{Ay} = 183{,}60\ kN$

$\Sigma M_A = 0 \Rightarrow R_{By} \cdot 13 - 40 \frac{3}{2} \cdot \left(\frac{3}{3} + 10\right) - 40 \cdot 6 \cdot \left(4 + \frac{6}{2}\right) - \frac{40 \cdot 4}{2} \cdot \left(\frac{2}{3} \cdot 4\right) = 0$

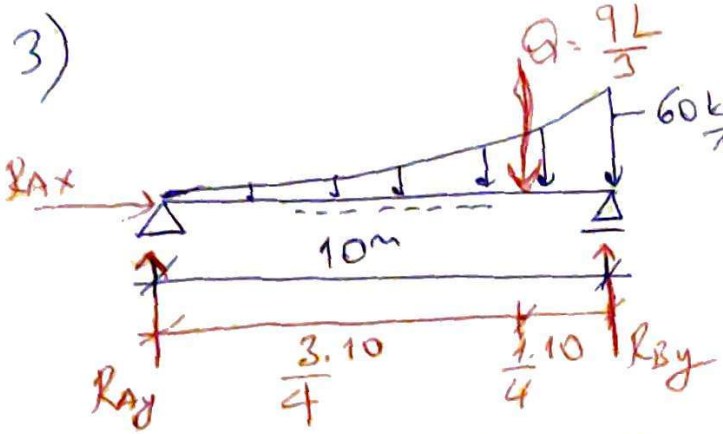
$R_{By} = 196{,}40\ kN$

Check! $\Sigma P_y = 0$

$40 \frac{4}{2} + 40 \cdot 6 + 40 \frac{3}{2} = 183{,}60 + 196{,}40$

$380 = 380$

3)



Note: $q(x) = \frac{9}{L^2} \cdot (L-x)^2$

$\Sigma P_x = 0 \Rightarrow R_{Ax} = 0$

$\Sigma M_B = 0 \Rightarrow R_{Ay} \cdot 10 - \frac{60 \cdot 10}{3} \cdot \frac{10}{4} = 0 \Rightarrow R_{Ay} = 50\,kN$

$\Sigma M_A = 0 \Rightarrow R_{By} \cdot 10 - \frac{60 \cdot 10}{3} \cdot \frac{3}{4} 10 = 0 \Rightarrow R_{By} = 150\,kN$

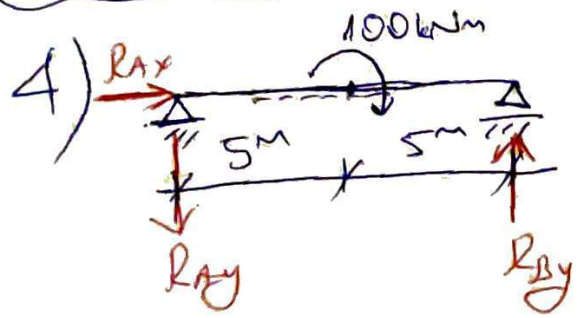
Check: $\Sigma P_y = 0 \Rightarrow R_{Ay} + R_{By} = Q$

$50 + 150 = \frac{60 \cdot 10}{3}$

$200 = 200$

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4)



$\Sigma P_x = 0 \Rightarrow R_{Ax} = 0$

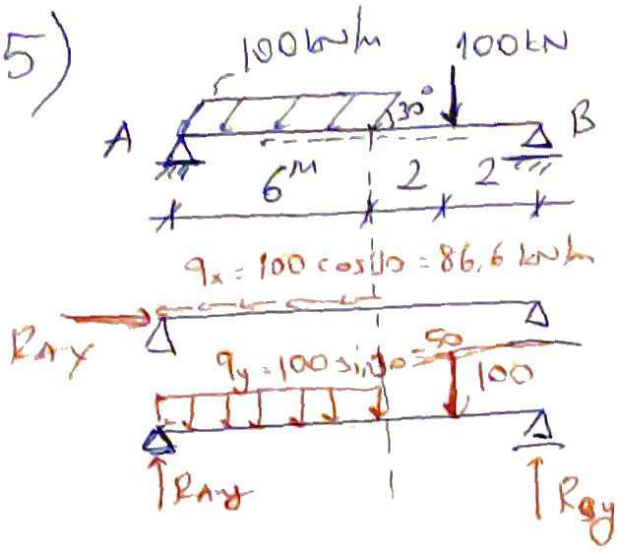
$\Sigma M_B = 0 \Rightarrow R_{Ay} \cdot 10 - 100 = 0$

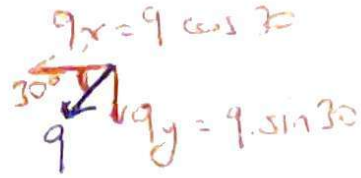
$R_{Ay} = 10\,kN$

$\Sigma M_A = 0 \Rightarrow R_{By} \cdot 10 - 100 = 0$

$R_{By} = 10\,kN$

5)





$\Sigma P_x = 0$

$R_{Ax} = 86,6 \cdot 6 = 519,6$ kN

$\Sigma M_B = 0 \Rightarrow R_{Ay}.10 - 50.6.\left(\frac{6}{2}+4\right) - 100.2 = 0$

$R_{Ay} = 230$ kN

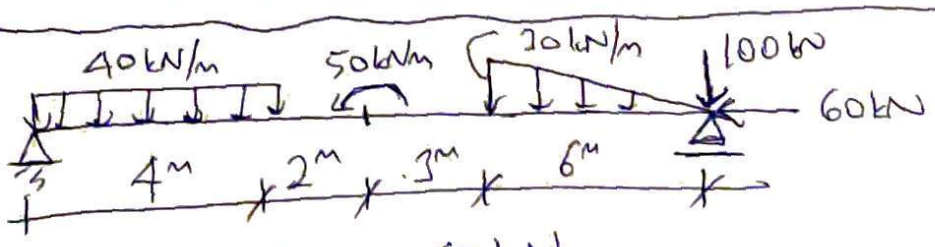
$\Sigma M_A = 0 \Rightarrow R_{By}.10 - 100.8 - 50.6.\left(\frac{6}{2}\right) = 0$

$R_{By} = 170$ kN

$\Sigma P_y = 0 \Rightarrow 6.50 + 100 = R_{Ay} + R_{By}$

$400 = 400$

6)



$\Sigma P_x = 0 \Rightarrow R_{Ax} = 60$ kN

$\Sigma M_B = 0 \Rightarrow R_{Ay}.15 - 40.4.\left(\frac{4}{2}+11\right) - 50 - \frac{30.6}{2}.\left(\frac{2}{3}6\right) = 0$

$R_{Ay} = 166$ kN

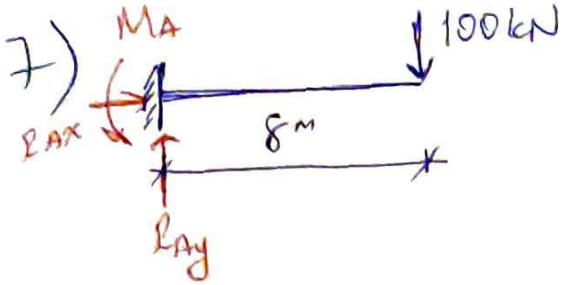
$\Sigma M_A = 0 \Rightarrow R_{By}.15 - 100.15 - \frac{30.6}{2}.\left(\frac{6}{3}+9\right) + 50 - 40.4.\frac{4}{2} = 0$

$R_{By} = 184$ kN

$\Sigma P_y = 0 \quad 40.4 + 30.\frac{6}{2} + 100 = R_{By} + R_{Ay}$

$350 = 350$

7)



$\Sigma P_x = 0 \Rightarrow R_{Ax} = 0$

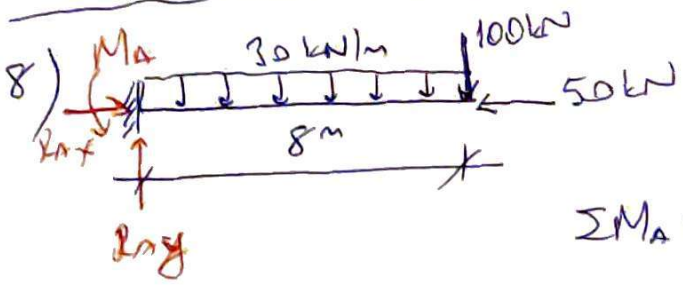
$\Sigma P_y = 0 \Rightarrow R_{Ay} = 100\,kN$

$\Sigma M_A = 0 \Rightarrow M_A - 100 \cdot 8 = 0$

$M_A = 800\,kNm$

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8)



$\Sigma P_x = 0 \Rightarrow R_{Ax} = 50\,kN$

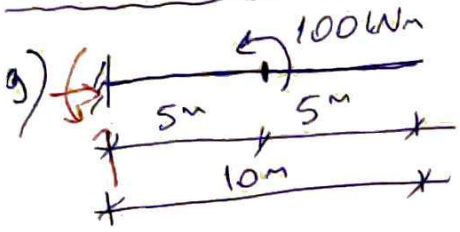
$\Sigma P_y = 0 \Rightarrow R_{Ay} = 30 \cdot 8 + 100 = 340\,kN$

$\Sigma M_A = 0 \Rightarrow M_A - 30 \cdot 8 \cdot 4 - 100 \cdot 8 = 0$

$M_A = 1760\,kNm$

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9)



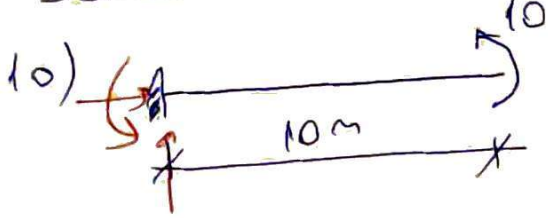
$\Sigma P_x = 0 \Rightarrow R_{Ax} = 0$

$\Sigma P_y = 0 \Rightarrow R_{Ay} = 0$

$\Sigma M_A = 0 \quad M_A = 100\,kNm$

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10)



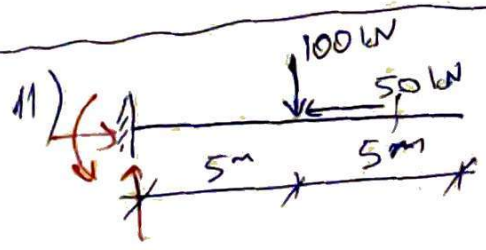
$\Sigma P_x = 0 \Rightarrow R_{Ax} = 0$

$\Sigma P_y = 0 \Rightarrow R_{Ay} = 0$

$\Sigma M_A = 0 \Rightarrow M_A = 100\,kNm$

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11)

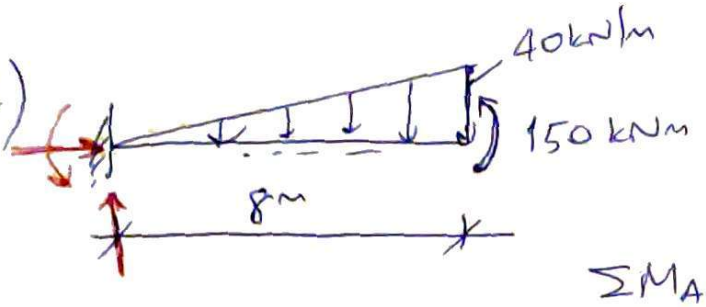


$\Sigma P_x = 0 \Rightarrow R_{Ax} = 50\,kN$

$\Sigma P_y = 0 \Rightarrow R_{Ay} = 100\,kN$

$\Sigma M_A = 0 \Rightarrow M_A = 100 \cdot 5 = 500\,kNm$

12)



$\Sigma P_x = 0 \Rightarrow R_{Ax} = 0$

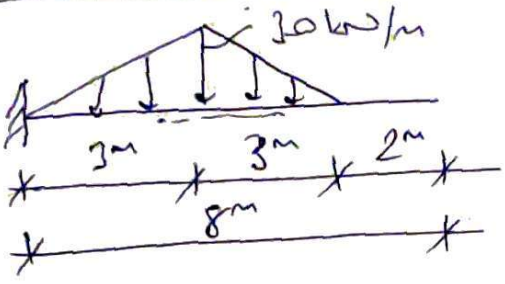
$\Sigma P_y = 0 \Rightarrow R_{Ay} = 40 \cdot \frac{8}{2} = 160 kN$

$\Sigma M_A = 0 \Rightarrow M_A = \frac{40 \cdot 8}{2} \cdot \frac{2}{3} \cdot 8 - 150$

$= 701{,}33 \, kNm$

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13)



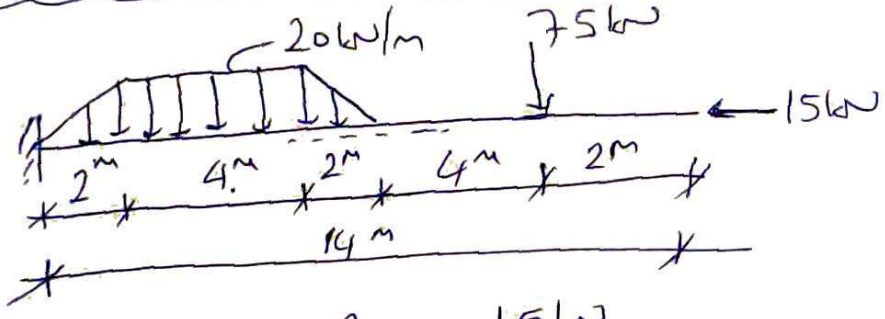
$\Sigma P_x = 0 \Rightarrow R_{Ax} = 0$

$\Sigma P_y = 0 \Rightarrow R_{Ay} = 30 \cdot \frac{6}{2} = 90 kN$

$\Sigma M_A = 0 \Rightarrow M_A = \frac{30 \cdot 6}{2} \cdot 3 = 270 kNm$

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14)

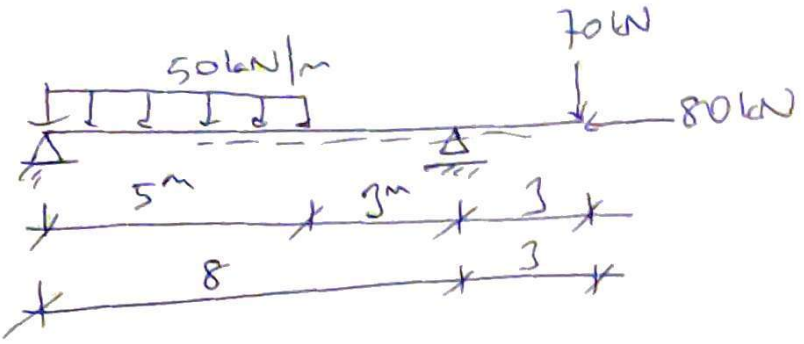


$\Sigma P_x = 0 \Rightarrow R_{Ax} = 15 kN$

$\Sigma P_y = 0 \Rightarrow R_{Ay} = \left(20 \cdot \frac{2}{2} + 20 \cdot 4 + 20 \cdot \frac{2}{2}\right) + 75 = 195 kN$

$\Sigma M_A = 0 \Rightarrow M_A = (120) \cdot 4 + 75 \cdot 12 = 1380 kNm$

15)



$\Sigma P_x = 0 \Rightarrow R_{Ax} = 80\,kN$

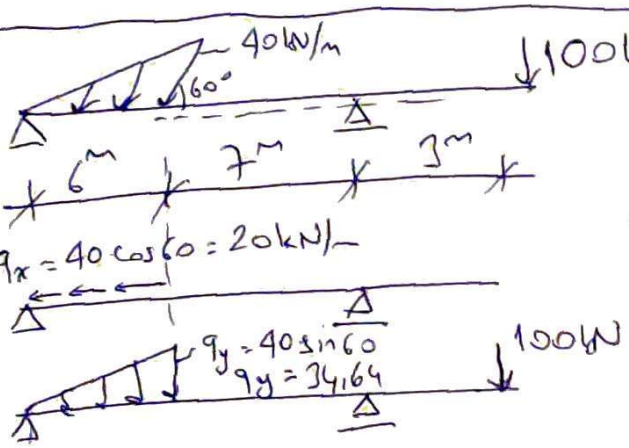
$\Sigma M_B = 0 \Rightarrow R_{Ay} \cdot 8 = 50 \cdot 5 \cdot \left(\frac{5}{2} + 3\right) - 70 \cdot 3$

$R_{Ay} = 145{,}62\,kN$

$\Sigma M_A = 0 \Rightarrow R_{Ax} = \dfrac{50 \cdot 5 \cdot 2{,}5 + 70 \cdot 11}{8} = 174{,}38\,kN$

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16)



$\Sigma P_x = 0 \Rightarrow R_{Ax} = 20 \cdot \frac{6}{2} = 60\,kN$

$\Sigma M_B = 0 \Rightarrow R_{Ay} = \dfrac{34{,}64 \cdot \frac{6}{2} \cdot \left(\frac{6}{3} + 7\right) - 100 \cdot 3}{13}$

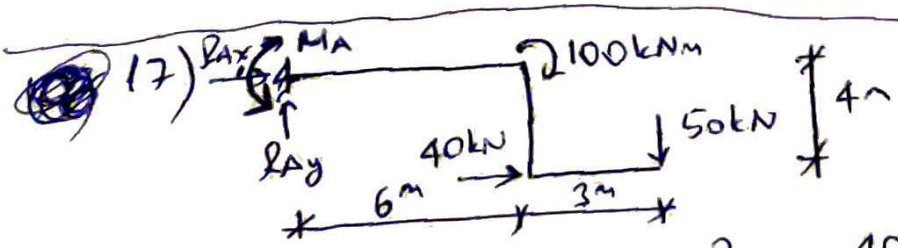
$= 48{,}87\,kN$

$\Sigma M_A = 0 \Rightarrow R_{By} = \dfrac{34{,}64 \cdot \frac{6}{3} \cdot \left(\frac{2}{3} \cdot 6\right) + 100 \cdot 16}{13}$

$R_{By} = 155{,}05\,kN$

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17)



$\Sigma P_x = 0 \Rightarrow R_{Ax} + 40 = 0 \Rightarrow R_{Ax} = -40\,kN$

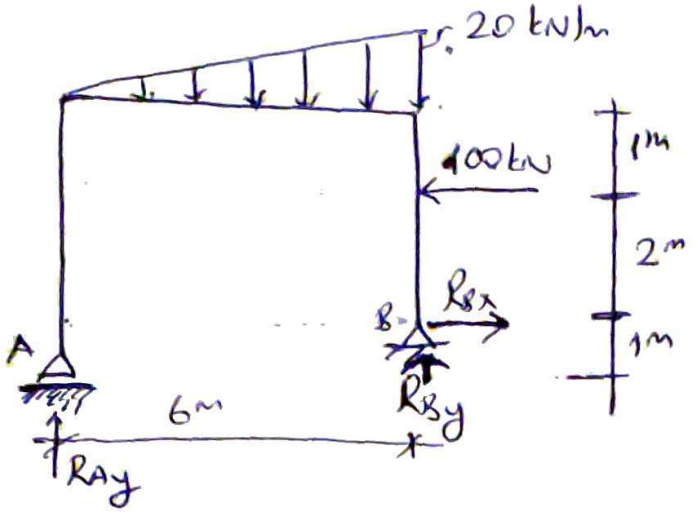
$\Sigma P_y = 0 \Rightarrow R_{Ay} - 50 = 0 \Rightarrow R_{Ay} = 50\,kN$

$\Sigma M_A = 0 \Rightarrow M_A - 100 + 40 \cdot 4 - 50 \cdot 9 = 0$

$M_A = 100 - 160 + 450$

$M_A = 390\,kNm$

18)



Estimate the support reactions of the given system.

$\Sigma P_x = 0 \Rightarrow R_{Bx} = 100\,kN$

$\Sigma M_B = 0 \Rightarrow R_{Ay} = \dfrac{20.6.0,5.2 + 100.2}{6} = 53,33\,kN$

$\Sigma P_y = 0 \Rightarrow R_{Ay} + R_{By} = 20.6.0,5$

$53,33 + R_{By} = 60\,kN \Rightarrow R_{By} = 6,67\,kN$

* 2nd way to estimate R_{By}:

Take moment respect to the intersection point of left column and support B, thus R_{Bx} can be eliminated in the equation

$$R_{By} = \frac{20.6.0,5.4 - 100.2}{6} = 6,67\,kN$$

* 3rd way to estimate R_{By}:

Take moment respect to support A.

$\Sigma M_A = 0$

$$R_{By} = \frac{20.6.0,5.4 - 100.3 + 100.1}{6} = 6,67\,kN$$